Inductive Bias under Data Scarcity: A Parameter-Matched Comparison of CNN, ViT, and CCT Architectures on CIFAR-10

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Abstract

Determining how architectural choices govern sample efficiency is a central challenge in scaling vision models to data-scarce regimes. We address this through a comparison of three architectures with matched parameter counts: a ResNet-style Convolutional Neural Network (CNN), a Vision Transformer (ViT), and a Compact Convolutional Transformer (CCT), trained on CIFAR-10 across five data fractions and two parameter regimes. Beyond accuracy, we examine internal representations using Centered Kernel Alignment (CKA), linear probing and mean attention distance. At matched 5M parameters, the CNN outperforms the ViT at every fraction, while at 0.75M parameters the CCT closes the gap, marginally surpassing it at full data. This advantage becomes even more pronounced in the absence of augmentation. Our central finding concerns data augmentation: contrary to the standard prediction that ViTs benefit most, the CNN gains substantially more from random crop and flip at low data, with the ranking reversing at higher fractions. We propose a mechanistic account: augmentation acts as a signal amplifier at low data and as a regularizer at high data. The CCT remains independent of augmentation throughout, as its convolutional tokenizer pre-encodes the relevant invariances.

1 Introduction

Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) encode fundamentally distinct inductive biases. While CNNs [1, 2] hard-wire locality and translation equivariance through shared convolutional filters, ViTs [3] delegate spatial reasoning entirely to data via global self-attention mechanisms. This structural difference becomes highly consequential under data scarcity, where ViTs typically require massive pre-training and struggle to match the sample efficiency of CNNs. Although Compact Convolutional Transformers (CCTs) [4] attempt to bridge this gap by introducing a convolutional tokenizer ahead of the attention stack, the precise contributions of locality, weight sharing, and global attention have not yet been fully disentangled in a controlled, parameter-matched setting.

To address this limitation, we investigate the data scale at which the vanilla ViT closes the performance gap with the CNN, and whether the hybrid CCT can outperform both baselines in extreme low-data regimes. Furthermore, we explore whether standard data augmentations benefit one architecture disproportionately across different data scales, and how internal representations differ across these three models. Our work provides three main contributions. First, we establish a parameter-matched benchmarking framework where we evaluate the models across five data fractions under 5M and 0.75M parameter budgets, specifically optimized to isolate the effects of pure CNNs, vanilla ViTs, and hybrid CCTs. Second, we deploy a multi-lens suite of analytical tools, including Centered Kernel Alignment (CKA), linear probing, and mean attention distance, to map how inductive biases dictate latent representations under data constraint. Finally, we uncover a counterintuitive low-data augmentation crossover, providing a mechanistic explanation for why conventional assumptions regarding augmentation reliance fail at small data scales.

2 Background and Related Work

Vision Transformers and Data Efficiency. Dosovitskiy et al. [3] demonstrated that vanilla ViTs require massive pre-training datasets to match the performance of CNNs, formalizing their inherent data reliance due to the lack of intrinsic spatial inductive biases. To address this limitation, Hassani et al. [4] proposed the CCT, which injects a spatial prior by replacing the standard patch projection with a shallow convolutional tokenizer and introducing a learned sequence pooling mechanism. While this approach successfully mitigates the data-hunger typical of attention models, existing literature does not systematically isolate the individual contributions of locality, weight sharing, and global attention under strict, parameter-matched capacity constraints.

Representation Analysis and Latent Geometry. To compare latent representations across fundamentally different architectures, Kornblith et al. [5] introduced Centered Kernel Alignment (CKA) as a robust similarity metric invariant to orthogonal transformations and isotropic scaling. Complementary to this, Alain & Bengio [6] formalized the use of linear probes to assess the linear separability of intermediate representations, providing a layer-wise view of feature quality and information relevant to the task. Raghu et al. [7] integrated CKA and attention distance to demonstrate that ViTs exploit global self-attention from the earliest layers, constructing features through a uniform geometry that is distinct from the local-to-global spatial hierarchy of CNNs. Unlike these prior works, which predominantly inspect representations in models trained on massive datasets, we deploy this integrated suite of diagnostics to track how internal representational geometry evolves under progressive, extreme data scarcity.

3 Method

Model Architectures. All candidate models are designed to process 32×32 RGB inputs and map them to 10-class logits, with total parameter counts strictly matched within a $\pm 5\%$ tolerance at each capacity budget. We denote the 0.75M parameter regime as small and the 5M parameter regime as large. The CNN baselines follow standard convolutional designs, with the large model implemented as a ResNet-style architecture and the small model as a lightweight max-pooling network. The vanilla ViT partitions images into non-overlapping patches, augments the resulting token sequence with a `CLS` token and learnable positional embeddings, and processes it through Pre-LN Transformer encoder blocks. The hybrid CCT replaces patch embedding with a convolutional tokenizer before applying Transformer blocks and a learned sequence-pooling mechanism. Full architectural specifications are reported in Table 1.

Training Protocol and Data Regimes. All models are trained for 150 epochs using an identical optimization protocol, ensuring that performance differences arise from architectural choices rather than training configurations. Full training hyperparameters are reported in Table 2. To evaluate behavior under data scarcity, each configuration is trained across five nested data fractions (10%, 25%, 50%, 75%, and 100%), where each smaller fraction is a strict subset of every larger fraction. This ensures that observed differences reflect data quantity rather than dataset identity. Training is performed under two augmentation regimes: a baseline condition without transformations and an augmented condition employing standard spatial perturbations. Final classification accuracy is evaluated under both conditions, whereas all representation diagnostics are computed exclusively on the augmented models. All experiments are conducted using a single fixed random seed.

4 Experiments

To characterize model behavior beyond classification accuracy, we complement the analysis with three representation diagnostics: linear CKA, linear probing, and mean attention distance. The complete implementation, trained checkpoints, and analysis scripts are publicly available in the accompanying repository [8]. Combined, these methodologies provide a comprehensive view of representational similarity, feature separability, and attention locality across the evaluated architectures.

4.1 Results

Accuracy Across Data Scales. Table 3a reports the final test accuracy at epoch 150 under the augmented training condition. In the 5M parameter regime, the CNN baseline leads at every data fraction, with the CCT tracking closely behind and the vanilla ViT trailing by a wide margin across all experimental conditions. Although the ViT improves markedly as more data become available, reducing the gap from roughly 25 percentage points at 10% data to 9 points at full data, it never reaches CNN performance and remains the weakest model. In the compact 0.75M regime, the advantage of the CNN gradually diminishes as the available data volume increases; here, the hybrid CCT matches and marginally surpasses the convolutional baseline at full data (89.65% vs. 89.27%). Without augmentation (Table 3b), the same ranking is observed in the 5M regime. In contrast, the low-capacity regime shifts noticeably in favour of the CCT, which overtakes the CNN already at 25% data and widens the gap thereafter, reaching 86.70% versus 81.73% at full data.

Centered Kernel Alignment and Representational Geometry. Cross-architecture CKA at the final layer reveals a capacity-dependent shift in representational similarity (Figures 1, 2). In the 5M regime, the CNN–CCT pair exhibits the highest similarity (approximately 0.65 across all data fractions), indicating that the convolutional tokenizer aligns CCT representations closely with those of the CNN. By contrast, the CNN–ViT pair remains the most divergent, while the ViT–CCT similarity increases steadily with data, rising from 0.47 at 10% to 0.70 at full data. A different pattern emerges in the 0.75M regime. Here, the CNN–CCT similarity drops substantially (approximately 0.32–0.35), becoming the least similar architecture pair, whereas CNN–ViT similarity remains comparatively higher (around 0.52–0.58). As in the larger regime, the ViT–CCT similarity increases with data, growing from 0.40 at 10% to 0.64 at full data. Taken together, these results suggest that the convolutional tokenizer exerts a stronger influence on the CCT representation geometry at larger capacity, where it pulls the model closer to the CNN. Under tighter parameter budgets, this alignment weakens, while the shared Transformer backbone becomes increasingly reflected in the growing similarity between ViT and CCT representations.

Linear Probing Dynamics. Probe accuracy across individual layers in Figure 3 reveals qualitatively distinct depth profiles for the three architectures. The CCT achieves remarkably high probe accuracy already at depth 0 (approximately 65%–75% across fractions and capacity regimes), indicating that its convolutional tokenizer extracts highly informative features before any Transformer processing. In contrast, the CNN displays the steepest rise across depth, consistent with a progressive hierarchical feature construction, while ViT remains comparatively flat across all data fractions with useful representations emerging only in the final blocks.
These trends are summarized by the final-layer probe accuracies shown in Figure 4. The vanilla ViT consistently produces the least linearly separable representations, trailing the best-performing model by 16-27 percentage points in the 5M regime and by 13-19 points in the 0.75M regime. The CCT achieves the most linearly separable final representations at low capacity, overtaking the CNN from 25% data onwards and reaching 89.2% probe accuracy at full data compared to 81.9% for the CNN. Notably, the small CCT at full data almost matches the large CNN at only 25% data (88.6%). Overall, the probe analysis suggests that the convolutional tokenizer provides the CCT with a substantial representational advantage, particularly under tight parameter budgets.

Mean Attention Distance and Spatial Scale. Mean attention distance reveals distinct attention strategies across the two Transformer-based architectures. In Figure 5 we can see that at full training data, the vanilla ViT exhibits a pronounced mid-depth dip in attention distance, reaching approximately 12.9 pixels in the 5M regime and 13.2 pixels in the 0.75M regime, indicating a temporary shift toward localized attention before returning to more global ranges in the final layers. In contrast, the CCT displays a smoother progression from local to global attention, increasing from roughly 14.2 to 16.4 pixels at 5M and remaining comparatively stable (15.3-15.9 px) at 0.75M. The per-head analysis in Figure 6 further highlights a marked difference in attention diversity. The ViT develops a substantially wider spread of attention distances, with some heads exceeding 20 px while others remain strongly local. By contrast, CCT heads remain concentrated within a much narrower 13-17 px range. Together, these results suggest that the ViT relies on a heterogeneous mixture of local and global attention patterns, whereas the CCT maintains a more consistent spatial hierarchy throughout the network.

4.2 Ablations

The Augmentation Crossover. Every candidate model was re-trained with data augmentation completely disabled to isolate the net augmentation gain compiled in Table 3c. Here, a counterintuitive pattern is revealed: the conventionally expected ranking (ViT $>$ CNN $>$ CCT), based on the theoretical premise that augmentation compensates for missing spatial inductive biases, is reversed under severe data scarcity. At the 10% data fraction, the CNN benefits most from augmentation, gaining 12.3 percentage points in the 5M regime and 9.3 points in the 0.75M regime, whereas the vanilla ViT gains only about 3.4 points. This ranking reverses as more data become available: ViT gains exceed 10 points at higher fractions, while CNN gains steadily decline. In contrast, the CCT remains largely augmentation-independent, maintaining gains within a narrow 2-4 point range across all data fractions and parameter budgets.

Mechanistic Account. One possible interpretation of these dynamics is that augmentation plays different functional roles across data regimes. Under severe data scarcity, augmentation may act primarily as a signal amplifier for the CNN: weight sharing allows each augmented sample to reinforce useful spatial invariances across many locations simultaneously. By contrast, the vanilla ViT lacks this built-in prior and cannot yet effectively absorb the structural diversity introduced by augmentation, leaving its parameter gradients heavily dominated by noise. As data availability increases, this dynamic appears to reverse. The ViT becomes increasingly capable of leveraging augmentation as a regularizer against positional overfitting, while the CNN experiences diminishing returns. The relatively stable behavior of the CCT is consistent with the idea that its convolutional tokenizer already encodes much of the relevant spatial structure before the attention blocks are applied.

Parameter Budget Ablation. Scaling down the capacity budget from 5M to 0.75M reduces final test accuracy on average by 4.8 pp for the CNN, 2.6 pp for the ViT, and only 1.8 pp for the CCT. The hybrid architecture therefore exhibits the strongest resilience to capacity reduction. A plausible explanation is that the convolutional tokenizer injects useful spatial structure at a relatively fixed parametric cost, making the CCT less dependent on scaling the Transformer backbone than a pure ViT. This observation is consistent with prior findings on the sample efficiency of compact convolutional transformers [4].

5 Conclusion

In this work, we have systematically demonstrated that the CNN baseline dominates the vanilla ViT under strictly matched capacity at every training data fraction, while the hybrid CCT successfully closes this performance gap and marginally overtakes the CNN under low capacity budgets at full data. This advantage becomes considerably stronger in the absence of augmentation. Representation analyses based on CKA, linear probing, and mean attention distance revealed that the three architectures develop distinct representational geometries, which become only partially aligned as data availability increases. Our central finding is a counterintuitive augmentation crossover: CNNs benefit most from augmentation under severe data scarcity, whereas ViTs benefit more as data availability increases. The CCT, by contrast, remains comparatively insensitive to augmentation across all conditions. Despite these insights, certain limitations remain. All experiments were conducted with a single random seed, preventing a statistical assessment of performance variability, and CIFAR-10 may not fully reflect behavior on larger-scale vision tasks. Future work could therefore extend this analysis through multi-seed evaluation, larger datasets, and alternative hybrid architectures.

References

[1] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.

[2] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.

[3] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.

[4] Ali Hassani, Steven Walton, Nikhil Shah, Abulikemu Abuduweili, Jiachen Li, and Humphrey Shi. Escaping the big data paradigm with compact transformers. *arXiv preprint arXiv:2104.05704*, 2021.

[5] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International conference on machine learning*, pages 3519–3529. PMlR, 2019.

[6] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*, 2016.

[7] Maithra Raghu, Thomas Unterthiner, Simon Kornblith, Chiyuan Zhang, and Alexey Dosovitskiy. Do vision transformers see like convolutional neural networks? *Advances in neural information processing systems*, 34:12116–12128, 2021.

[8] Victor Antonescu, Letizia Di Pietro, Martin Lukanov, Alexandru Stoica, and Antonio Troiano. Inductive bias under data scarcity: Code repository. `https://github.com/trozki213/cnn-vit-cct-comparison`, 2026. GitHub repository.

A Additional Results and Implementation Details

A.1 Architecture details

Table 1: Architecture hyperparameters and trainable parameter counts.

	CNN_small	ViT_small	CCT_small	CNN_large	ViT_large	CCT_large
Params	765,546	760,906	727,467	4,904,522	4,976,970	5,261,467
Patch / token size	—	4×4	conv 3×3	—	4×4	conv 3×3
Embed dim	—	128	160	—	320	300
Depth (blocks)	3 stages	5	4	3 stages	6	7
Attention heads	—	4	4	—	4	4
MLP dim	—	320	160	—	640	600
Pooling	MaxPool×3	CLS	SeqPool	AdaptiveAvg	CLS	SeqPool
Dropout	0.5	0.1	0.1	0.2	0.1	0.1

Table 2: Training hyperparameters (identical across all six models and all data fractions).

Hyperparameter	Value
Epochs	150
Optimiser	AdamW
β_1, β_2	0.9, 0.999
Peak learning rate	1×10^{-3}
Weight decay	5×10^{-2}
LR schedule	Linear warm-up ($\approx$ 7 ep.) $\rightarrow$ cosine annealing to 0
Batch size	256
Loss	Cross-entropy, label smoothing $\epsilon = 0.1$
Gradient clipping	$\|g\|_2 \leq 1.0$
Mixed precision	AMP (GPU only)
Random seed	0 (single-seed)
Hardware	NVIDIA GPU (training); CPU/MPS (analysis)

A.2 Accuracy Results

Table 3: Accuracy and augmentation gain at epoch 150.

(a) Test accuracy (%) with augmentation

Model	10%	25%	50%	75%	100%
CNN_large	**79.80**	**88.57**	**92.18**	**93.94**	**94.91**
ViT_large	54.40	68.28	77.53	82.43	85.80
CCT_large	70.47	81.13	86.55	89.36	90.84
CNN_small	**77.40**	**83.41**	**87.07**	**88.28**	89.27
ViT_small	53.73	65.52	74.08	79.58	82.60
CCT_small	68.11	79.19	84.90	87.62	**89.65**

(b) Test accuracy (%) without augmentation

Model	10%	25%	50%	75%	100%
CNN_large	**67.50**	**78.61**	**84.74**	**87.66**	**90.03**
ViT_large	50.96	58.81	67.41	71.92	76.19
CCT_large	66.19	77.24	83.76	86.94	88.79
CNN_small	**68.10**	74.34	78.23	80.24	81.73
ViT_small	50.27	58.10	64.95	69.45	73.15
CCT_small	65.18	**76.05**	**81.81**	**84.96**	**86.70**

(c) Augmentation gain (pp) = (a) − (b)

Model	10%	25%	50%	75%	100%
CNN_large	**12.30**	9.96	7.44	6.28	4.88
ViT_large	**3.44**	9.47	10.12	10.51	9.61
CCT_large	4.28	3.89	2.79	2.42	2.05
CNN_small	**9.30**	9.07	8.84	8.04	7.54
ViT_small	**3.46**	7.42	9.13	10.13	9.45
CCT_small	2.93	3.14	3.09	2.66	2.95

A.3 CKA: within-model and cross-architecture

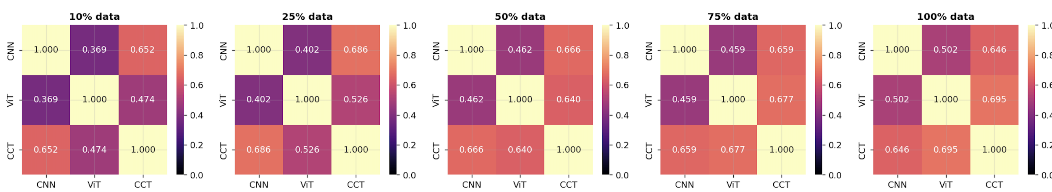

Figure 1: Final-layer cross-architecture CKA at the 5 M parameter regime, all five data fractions.

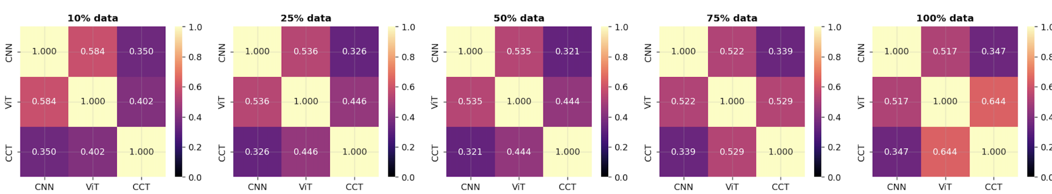

Figure 2: Final-layer cross-architecture CKA at the 0.75 M parameter regime, all five data fractions.

A.4 Linear probes: layer-wise feature separability

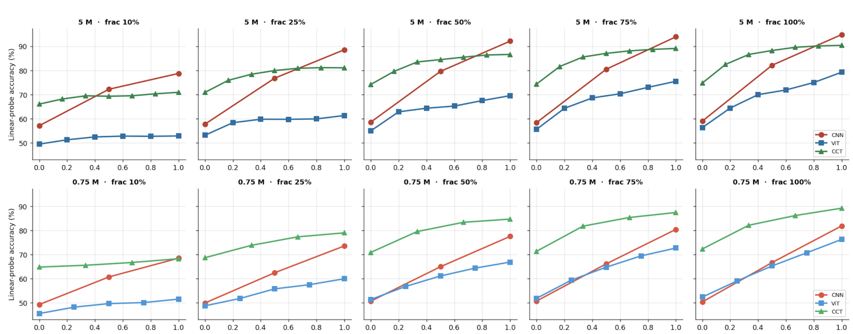

Figure 3: Layer-wise linear-probe accuracy vs. normalised depth, per regime (rows) and data fraction (columns).

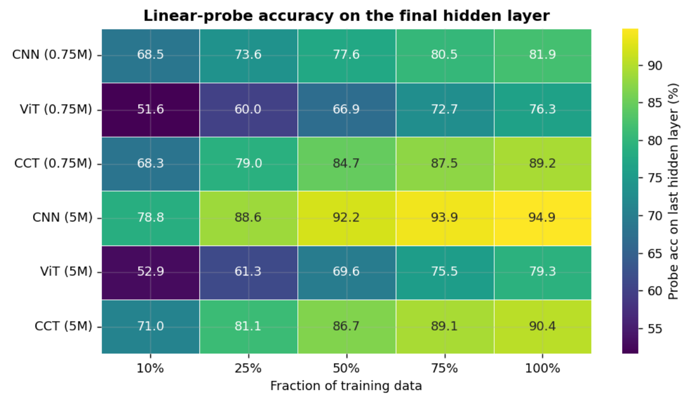

Figure 4: Final-hidden-layer linear-probe accuracy heatmap.

A.5 Mean attention distance: per-head detail

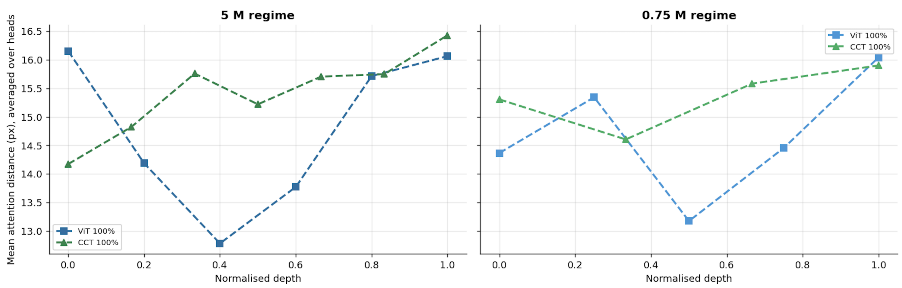

Figure 5: Mean attention distance vs. normalised depth (100% data).

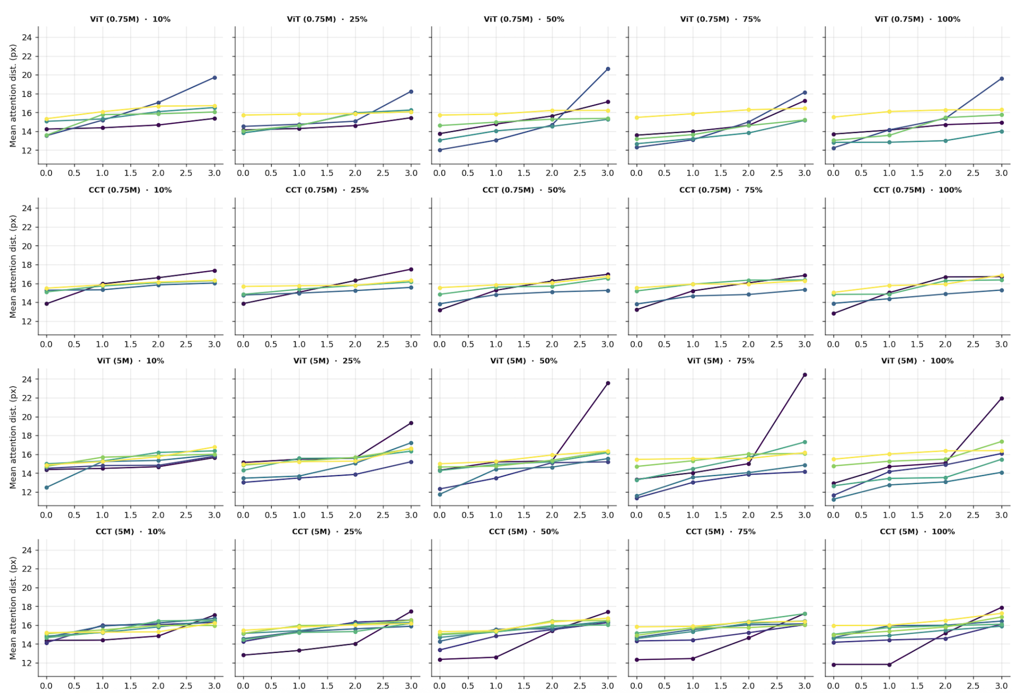

Figure 6: Mean attention distance per head per layer, sorted within each layer. Each dot is one head; line colour encodes layer depth (darker = earlier, lighter = later). Top two rows: 0.75 M; bottom two rows: 5 M.